

# 1 Introduction

In guitar, tone control is a major technical skill. Each musical instrument’s quality of sound—its timbre—is differentiated by the varying composition of harmonic frequencies ( $f_n = nf_1$  for  $n = 1, 2, 3...$ ), which are present in addition to the fundamental frequency  $f_1$ . In addition to the fundamental variation between instruments, the timbre also varies based on how notes are played (Perov et al., 2016). It is qualitatively observed that plucking near the bridge will create a harsher, “tinny” sound while plucking toward the bridge will create a fuller sound.

This investigation will explore how the quality of sound may change depending on where the string is plucked. The research question I have decided to investigate is: *How does changing the position of the plectrum along a guitar scale affect the frequency of the harmonic with the highest intensity when a string with constant tension is plucked?*

## 1.1 Background

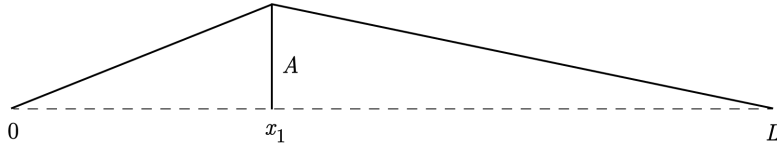


Figure 1: A guitar string of length  $L$  pulled a distance  $A$  at position  $x_1$

A guitar string is a string with two fixed ends. When the string is plucked (as in Figure 1), a transverse wave is initiated that propagates along the length of the string. The reflection of the transverse wave at the fixed ends superimposes with the original wave, forming nodes (points of zero displacement) and antinodes (points of maximum displacement). This creates a standing wave, which is a wave that does not propagate but rather oscillates in place (Tsokos, 2014).

The first harmonic is a standing wave with a single loop made of two nodes and one antinode, which results in a sound wave at the fundamental frequency, given by the equation  $f_1 = \frac{v}{2L}$ .

The linear principle of superposition states that the net displacement of a system subjected to multiple independent disturbances (waves, in this case) is the algebraic sum of the

displacements due to each disturbance. If  $y_1(x, t)$  and  $y_2(x, t)$  are two waveforms propagating along the same string, then the net waveform  $y(x, t)$  is given by the equation

$$y(x, t) = y_1(x, t) + y_2(x, t) \quad (1)$$

Due to this principle, multiple, independent standing waves are formed on the guitar string, which results in higher-order harmonics. Each subsequent harmonic is an integer multiple of the fundamental frequency (known as frequency quantization), given by

$$f_n = n f_1 \quad (2)$$

These individual waveforms superimpose linearly to form a composite waveform which is the sum of all individual harmonic waveforms. As a result, the sound emitted by a guitar string is a mixture of multiple harmonic frequencies.

To investigate how the picking length affects the harmonic with the highest amplitude, we must determine the variation of the amplitude of each harmonic frequency separately. The amplitude  $b_n$  of the  $n$ th harmonic is given by the equation

$$b_n = \frac{2A \sin\left(n\pi \frac{x_1}{L}\right)}{\frac{x_1}{L} \left(1 - \frac{x_1}{L}\right) \pi^2 n^2} \quad (3)$$

where  $A$  is the distance the string is pulled when plucked and  $L$  is the string length (Perov et al., 2016).

By calculating  $b_n$  (Equation 3) for  $n = 2...10$  at each variation of  $x_1$  and choosing the value of  $n$  that maximizes  $b_n$ , the non-fundamental harmonic frequency with the highest amplitude can be calculated using Equation 2.

Plotting the picking length  $x_1$  against the non-fundamental harmonic frequency with the highest amplitude  $f$  results in a Gaussian-like function where the 2<sup>nd</sup> harmonic has the highest amplitude across picking lengths, except near the middle of the fretboard where the 3<sup>rd</sup> harmonic has the highest amplitude.

Therefore, it is hypothesized that the non-fundamental harmonic frequency with the highest amplitude will remain constant across picking lengths  $x_1$ , except with a peak around the middle of the fretboard ( $x_1 = \frac{L}{2}$ ).

## 2 Methodology

### 2.1 Variables

| Type                 | Variable                                                                                             | Specification                                                                                                                                                                                                                                                                                                                                 |
|----------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Independent Variable | Picking length $x_1$ (m),<br>the distance in meters<br>from the nut where the<br>string is disturbed | $x_1 = \{0.01\text{m}, 0.08\text{m}, 0.15\text{m}, 0.22\text{m}, 0.29\text{m},$<br>$0.36\text{m}, 0.43\text{m}, 0.50\text{m}, 0.57\text{m}, 0.64\text{m}\},$<br>varied by plucking at the picking length as<br>indicated by a tape measure                                                                                                    |
| Dependant Variable   | Frequency $f$ of the non-<br>fundamental harmonic<br>with the highest<br>amplitude (Hz)              | Measured by recording the frequency $f_n$<br>of the $n$ th harmonic that has the highest<br>amplitude through FFT (Fast Fourier<br>Transform) analysis, where $n = 2...10$                                                                                                                                                                    |
| Controlled Variable  | Diameter, tension, age,<br>and other material<br>properties of the string                            | The same high E string tuned to the<br>standard $E = 329.63$ Hz will be used for all<br>the experiments because different material<br>properties may change the harmonic<br>composition. By extension, the density<br>will be kept constant by performing data<br>collection in the same air-conditioned room<br>(set to $23^\circ\text{C}$ ) |
|                      | Guitar body and<br>electronics                                                                       | The neck pickup on the same guitar will be<br>used across trials to minimize random error<br>from the frequency response imparted by<br>the pickups and guitar body.                                                                                                                                                                          |
|                      | Plectrum choice and<br>picking technique                                                             | The same pick will be used across trials.<br>The string will be plucked at a $45^\circ$ angle<br>to the fretboard, with the tip of the pick<br>contacting the fretboard. This controls the<br>variable $A$ in Equation 3, leaving just $x_1$<br>the independent variable.                                                                     |

Table 1: Variables

## 2.2 Apparatus & Experimental Setup

- Electric guitar (6-string *Kiesel Guitars* “*Osiris*” model used)
- 1 set of 0.009 – 0.042mm gauge nylon electric guitar strings (*D’Addario XS Nickel XSE0942 Super Light Coated Electric Guitar Strings* model used)
- 1 audio interface with instrument input (*Audient EVO4* used)
- 1 instrument cable
- 1 Computer with audio analysis and spreadsheet software (*MATLAB* and *Google Sheets* used)
- 1 tape measure
- 1 clamp stand
- Masking tape

Figure 2 presents the experimental setup used to collect data. The guitar is clamped down for ease of plucking, and the tape measure is placed directly on the 2nd and 3rd strings to be on the same axis as the 1st string (which is plucked).



Figure 2: Labeled photograph of experimental setup

## 2.3 Experimental Procedure

### PREPARATION

1. Clamp the guitar’s head using the clamp stand to hold it in place.
2. Lay the tape measure out along the fretboard on top of the 2nd and 3rd strings, securing it in place with the clamp stand and masking tape.
3. Connect the guitar to the audio interface using the instrument cable, and connect the audio interface to the computer.
4. Turn all knobs on the guitar to 10 (maximum position clockwise) and adjust the gain on the audio interface appropriately to prevent clipping of the audio signal.

## RECORDING AND ANALYSIS

5. Start the 5-second recording on the computer.
6. At the first variation of  $x_1$ , pluck the 1st string using the pick at a  $45^\circ$  angle to the fretboard, touching the pick to the fretboard and pushing towards the string perpendicularly.
7. Wait for the recording period to end and ensure the data is saved.
8. Using the frequency analysis software, convert the time-domain data into frequency-domain data.
9. Identify the true fundamental frequency by searching for a peak in the data within 5% of the expected fundamental frequency  $f_1 = 329.63$  Hz
10. Calculate the frequency of each non-fundamental harmonic  $n$  for  $n = 2 \dots 10$
11. Calculate the absolute amplitude  $|b_n|$  for the frequency at each harmonic, recording the frequency  $f_n$  that maximizes the absolute amplitude  $|b_n|$ .
12. Repeat steps 5–11 four more times to complete five trials for the variation.
13. Repeat steps 5–12 for the remaining variations, modifying the picking length  $x_1$  according to Table 1.

## PROCESSING

14. For each picking length  $x_1$ , calculate the average frequency  $f$  with the highest amplitude using a spreadsheet tool.
15. For each picking length  $x_1$ , calculate the average uncertainty in the frequency using 
$$\Delta f = \frac{f_{\max} - f_{\min}}{2}.$$

In data collection, steps 8-11 were automated using a custom MATLAB (scientific programming platform) program. The program applies a Hann window to the recorded audio data to reduce noise interference and computes the Fourier transform of the windowed data using Welch's method. The fundamental is identified by looking for peaks around the expected frequency, allowing for variations in tuning.

## 2.4 Safety, Ethical, and Environmental Considerations

This is a relatively safe experiment. As the guitar is directly connected to the audio interface, any voltages are low and are not likely to pose a safety risk. There is no risk of

water damage to any electronics. New strings are not required for the experiment and no consumable materials are used, limiting this experiment's environmental impacts.

### 3 Analysis

#### 3.1 Raw Data - Time Domain

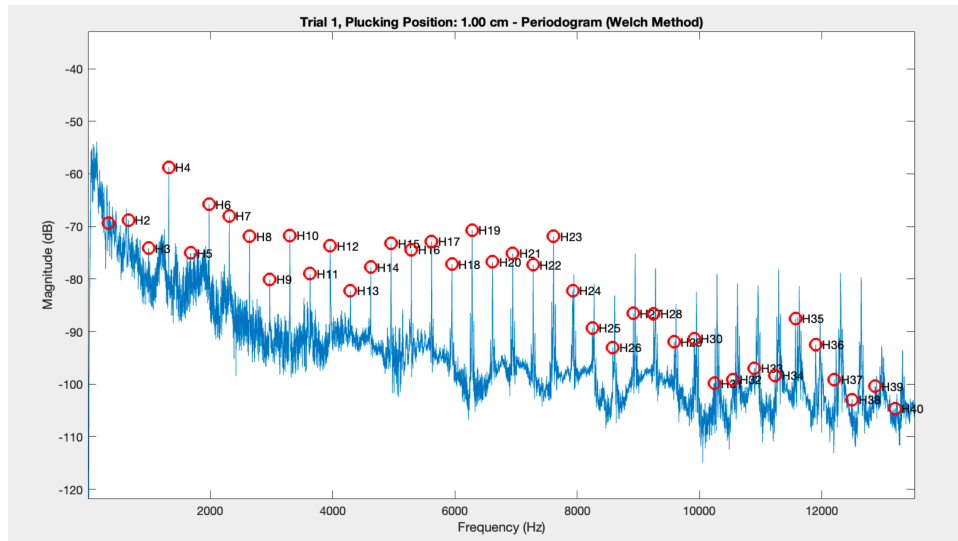


Figure 3: Example of frequency-domain data showing harmonic peaks

As an example, Figure 3 shows ...

After FFT,

#### QUALITATIVE OBSERVATIONS

It was observed that

### 3.2 Raw Data - Frequency Domain

| Plucking length                  | Frequency with highest amplitude |                     |         |         |         |            |
|----------------------------------|----------------------------------|---------------------|---------|---------|---------|------------|
| $x_1$ (m)                        | $f / \Delta f$ (Hz)              |                     |         |         |         |            |
| $\Delta x_1 = \pm 0.005\text{m}$ | Trial 1                          | Trial 2             | Trial 3 | Trial 4 | Trial 5 | $\Delta f$ |
| 0.01                             | 655.70                           | 655.88              | 655.88  | 655.88  | 655.88  | 32.96      |
| 0.08                             | 655.70                           | 655.88              | 655.88  | 655.88  | 655.88  | 32.96      |
| 0.15                             | 655.88                           | 655.88              | 655.88  | 656.07  | 655.88  | 32.96      |
| 0.22                             | 656.07                           | 655.88              | 655.88  | 656.07  | 656.07  | 32.96      |
| 0.29                             | 984.01                           | 983.83              | 984.01  | 984.01  | 984.01  | 49.44      |
| 0.36                             | 656.07                           | 984.01 <sup>1</sup> | 655.88  | 655.88  | 655.88  | 32.96      |
| 0.43                             | 655.88                           | 656.07              | 655.88  | 655.88  | 655.88  | 32.96      |
| 0.50                             | 655.88                           | 655.88              | 655.88  | 655.88  | 655.88  | 32.96      |
| 0.57                             | 656.07                           | 655.88              | 655.88  | 655.88  | 655.88  | 32.96      |
| 0.64                             | 655.88                           | 656.07              | 656.07  | 655.88  | 656.07  | 32.96      |

Table 2: Raw data of highest amplitude harmonic frequencies at each plucking length

#### ABSOLUTE UNCERTAINTIES

- The absolute uncertainty in the picking length  $x_1$  is  $\Delta x_1 = \pm 0.005\text{m}$  as the tape measure has a precision of 1 cm.
- The absolute uncertainty in the frequency  $f$  is

$$\Delta f = \pm n \times 0.05 f_1 = \pm n \times 16.48\text{Hz} \quad (4)$$

as the frequency search window in the FFT is 5% of the expected fundamental frequency  $f_1 = 329.63\text{Hz}$ . This uncertainty is multiplied by  $n$  because it propagates when the non-fundamental harmonic is calculated by multiplying the fundamental frequency by the harmonic number  $n$ . For instance, plucking length  $x_1 = 0.01\text{m}$  results in the 2nd harmonic with the highest amplitude, so regardless of the FFT-found fundamental frequency, the absolute uncertainty is  $\Delta f = \pm 2 \times 16.48 = \pm 32.96\text{Hz}$ .

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<sup>1</sup>Outlier removed for processed data

### 3.3 Processed Data

| Plucking length<br>$x_1$ (m)<br>$\Delta x_1 = \pm 0.005\text{m}$ | Average frequency with<br>highest amplitude<br>$\bar{f}$ (Hz) | Uncertainty in $\bar{f}$<br>$\Delta \bar{f}$ (Hz) |
|------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------|
| 0.01                                                             | 655.85                                                        | 32.96                                             |
| 0.08                                                             | 655.85                                                        | 32.96                                             |
| 0.15                                                             | 655.92                                                        | 32.96                                             |
| 0.22                                                             | 655.99                                                        | 32.96                                             |
| 0.29                                                             | 983.97                                                        | 49.44                                             |
| 0.36                                                             | 655.93                                                        | 32.96                                             |
| 0.43                                                             | 655.92                                                        | 32.96                                             |
| 0.50                                                             | 655.88                                                        | 32.96                                             |
| 0.57                                                             | 655.92                                                        | 32.96                                             |
| 0.64                                                             | 655.99                                                        | 32.96                                             |

Table 3: Raw data of highest amplitude harmonic frequencies at each plucking length

#### CALCULATIONS

- The average frequency with the highest amplitude  $\bar{f}$  was calculated by averaging the frequencies with the highest amplitude across trials, for each variation of the picking length  $x_1$ . For example, for  $x_1 = 0.01\text{m}$ , the average frequency  $\bar{f}$  is

$$\begin{aligned}
 \bar{f} &= \frac{1}{5} \sum_{n=0}^5 (f_{x_1=0.01, \text{trial } n}) \\
 &= \frac{1}{5} (655.70 + 655.88 + 655.88 + 655.88 + 655.88) \\
 &= 655.85 \text{ Hz}
 \end{aligned} \tag{5}$$

- Because the uncertainties are not from random error between trials but the FFT process, the absolute uncertainty is used, from Table 2. For instance, the absolute uncertainty for picking length  $x_1 = 0.01\text{m}$  is 32.96Hz, so  $\Delta \bar{f} = 32.96\text{Hz}$ .

### **3.4 Graphical Analysis**

## **4 Conclusion**

### **4.1 Scientific Reasoning**

## **5 Evaluation**

### **5.1 Strengths**

### **5.2 Limitations**

### **5.3 Further Study**

## References

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